

Supplementary Material for CIP-Net: Continual Interpretable Prototype-based Network

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Additional Experimental Setup Details

We used ConvNeXt-tiny (Liu et al. 2022a) as backbone. We performed a hyperparameter search for the learning rates, λ_R, λ_H . We used AdamW (Zhuang et al. 2022) optimizers with the learning rates and respective settings shown in Table 1. We set a batch size of 64 for all task settings and datasets apart for the 20-task setting on the CUB-200-2011 dataset, where the batch size used was 32. However, each mini-batch contains two views of an image, resulting in twice the dimension (batch sizes of 128 and 64). Input image dimensions were $224 \times 224 \times 3$. The backbone was pretrained on ImageNet and the classifiers' weights are initialized with values drawn from the normal distribution with mean 1.0 and standard deviation 0.1. During pretraining, λ_A is slowly increased to 1 (to prevent the trivial solution that every patch gets the same encoding), $\lambda_T = 1.0$, $\lambda_R = 8.0$. During training $\lambda_A = 5.0$, $\lambda_T = 1.0$, $\lambda_R = 8.0$, $\lambda_H = 10.0$, $\lambda_D = 0.005$, $\lambda_C = 2.0$. For the first epoch of training of each task, backbone f is frozen and only weights $\mathbf{W}^i$ of the current classification head are trained. The predictions during training follow the CIL scenario: the output of each classification head is concatenated and passed to the final softmax to get the prediction. During inference, prototype presence scores $p < 0.1$ are ignored.

Each model was trained in parallel on Tesla V100 GPUS.

Task-specific Results on CUB

In Table 3 we show detailed average accuracies for each task in the 4-task setting on CUB-200-2011. CIP-Net outperforms all the baselines. A bias towards the last task can be noticed in the CIL scenario, as commonly happens to CL models.

Our improvements mainly stem from reduced forgetting rather than backbone capacity. Additional preliminary experiments with ResNet34 as backbone also show that CIP-Net outperformed ICICLE in 4-task setting with the same hyperparameters used in the main experiments using ConvNeXt-tiny.

Results on Stanford Cars

Table 2 depicts how CIP-Net works on the Stanford Cars dataset compared to other baseline methods. Results are consistent with the ones on CUB and show that CIP-Net outperforms all the baselines considered.

Prototype Activation Drift

In Figure 2, we show other regularized prototypes that remain stable throughout training in the 4-task setting. While very rare, in Figure 2c, we show a regularized prototype that drifted in the 4-task setting. We also present an example of prototype activation drift in subfigure 2b, where a prototype was activated on a background region of the input image. Regarding this last case, it's important to emphasize that CIPNet's prediction relies more on the combination of activated prototypes rather than the presence of any single one. As such, prototypes should be visualized collectively to fully understand an explanation.

Local Explanations

For completeness in Figure 1, we show an example of local explanations for a prediction.

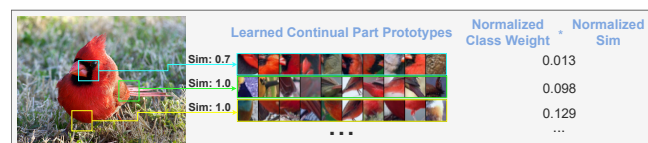


Figure 1: Example of local explanation of CIP-Net for the correct class. CIP-Net learns part-prototypes visualized as patches from the training data, and localizes similar image patches in an unseen input image.

Data Augmentation

For the CUB-200-2011 dataset, we first augment the dataset following ProtoPNet (Chen et al. 2019) pipeline¹ to have a fair comparison with ICICLE, then at runtime we use the data augmentation of PIP-Net (Nauta et al. 2023): we

¹<https://github.com/cfchen-duke/ProtoPNet>

use TrivialAugment (Müller and Hutter 2021) twice. TrivialAugment applies a single random transformation per image. The first application is for location-based transformations (shear, rotation, translation), the second for color-based ones (brightness, sharpness, hue, contrast). For the Stanford Cars dataset, only PIP-Net’s augmentation pipeline is applied.

Phase	Model layers	Learning rate	Scheduler	Duration
Pretraining	Backbone	$1 \cdot 10^{-4}$	Cosine Annealing	{10, 10, 10} epochs on CUB, {10, 7, 5} epochs on CARS
Training	Backbone	$1 \cdot 10^{-4}$	Cosine annealing	{20, 15, 10} epochs on CUB, {20, 13, 10} epochs on CARS
	τ	$1 \cdot 10^{-2}$		
	Classification heads	$1 \cdot 10^{-3}$	Cosine annealing with warm restart	

Table 1: CIP-Net’s hyperparameters

Method	Final Avg. Task-Incremental Accuracy			Final Avg. Class-Incremental Accuracy		
	4 Tasks	7 Tasks	14 Tasks	4 Tasks	7 Tasks	14 Tasks
EWC	0.456 ± 0.021	0.315 ± 0.037	0.287 ± 0.041	0.258 ± 0.019	0.152 ± 0.022	0.011 ± 0.009
LWM	0.459 ± 0.072	0.416 ± 0.048	0.305 ± 0.022	0.233 ± 0.026	0.171 ± 0.016	0.080 ± 0.008
LWF	0.375 ± 0.021	0.356 ± 0.024	0.250 ± 0.020	0.230 ± 0.011	0.171 ± 0.005	0.092 ± 0.008
ICICLE	0.654 ± 0.014	0.645 ± 0.003	0.583 ± 0.048	0.335 ± 0.005	0.203 ± 0.010	0.116 ± 0.018
CIP-Net	0.862 ± 0.008	0.921 ± 0.007	0.925 ± 0.012	0.589 ± 0.006	0.585 ± 0.009	0.389 ± 0.003
PIP-Net C	0.882 ± 0.005					

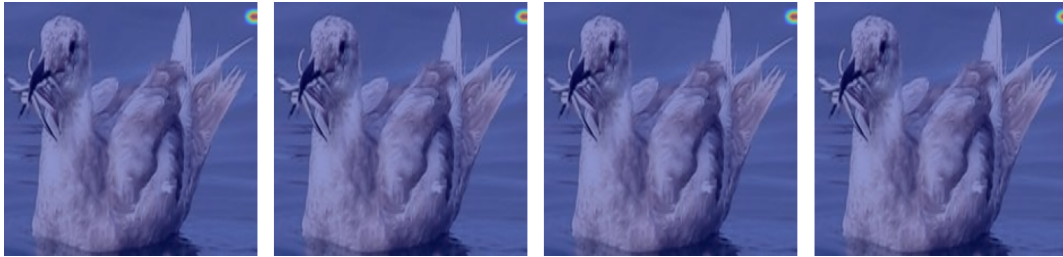
Table 2: Final average (3 seeds) accuracy comparison for different numbers of tasks on Stanford Cars, demonstrating the negative impact of the high number of tasks to be learned on models’ performance. Despite this trend, CIP-Net outperforms the baseline methods across all task numbers. At last, the PIP-Net (ConvNeXt) performance upper bound is presented. Results of methods different from CIP-Net were taken from Rymarczyk et al. (2023).

Method	Avg. Task-Incremental Accuracy					Avg. Class-Incremental Accuracy				
	Task 1	Task 2	Task 3	Task 4	Avg	Task 1	Task 2	Task 3	Task 4	Avg
EWC	0.24 ± 0.02	0.38 ± 0.07	0.54 ± 0.04	0.60 ± 0.05	0.44 ± 0.01	0.00 ± 0.00	0.06 ± 0.00	0.27 ± 0.03	0.53 ± 0.05	0.21 ± 0.01
LWF	0.17 ± 0.05	0.12 ± 0.01	0.23 ± 0.02	0.74 ± 0.06	0.30 ± 0.05	0.16 ± 0.03	0.00 ± 0.00	0.02 ± 0.00	0.54 ± 0.14	0.22 ± 0.02
LWM	0.19 ± 0.01	0.41 ± 0.01	0.43 ± 0.03	0.77 ± 0.01	0.45 ± 0.02	0.03 ± 0.02	0.02 ± 0.02	0.08 ± 0.01	0.77 ± 0.04	0.18 ± 0.03
ICICLE	0.52 ± 0.02	0.66 ± 0.05	0.71 ± 0.04	0.72 ± 0.00	0.65 ± 0.01	0.23 ± 0.01	<u>0.36 ± 0.02</u>	0.31 ± 0.01	0.49 ± 0.02	0.35 ± 0.05
CIP-Net (R)	0.74 ± 0.01	0.58 ± 0.11	0.66 ± 0.02	0.71 ± 0.07	<u>0.67 ± 0.01</u>	0.24 ± 0.01	0.25 ± 0.03	0.37 ± 0.07	0.61 ± 0.06	0.37 ± 0.01
CIP-Net (C)	0.83 ± 0.00	0.63 ± 0.16	0.81 ± 0.04	0.81 ± 0.07	0.77 ± 0.01	0.25 ± 0.05	0.42 ± 0.02	0.51 ± 0.04	0.70 ± 0.07	0.47 ± 0.01

Table 3: Average (3 seeds) task- and class-incremental accuracy across 4 tasks for each method on CUB-200-2011. (R) and (C) represent ResNet34 and ConvNeXt-tiny backbones respectively (Liu et al. 2022b). Results of methods different from CIP-Net were taken from Rymarczyk et al. (2023).



(a) Example of suppressed prototype activation drift for some of the regularized prototypes from task 1 to task 4 (4-task setting) on CUB-200-2011.



(b) Example of "background" prototype.



(c) A rare case of prototype activation drift in a regularized prototype.

Figure 2: (a) Examples of suppressed prototype activation drift for some of the regularized prototypes from task 1 to task 4 (4-task setting) on CUB-200-2011. (b) Example of prototype activation drift for a prototype that activated on the background of the input image. (c) Rare example of prototype activation drift in a regularized prototype.

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